



NERVOUS SYSTEM — FULL MIND MAP

Biology Ch. 7.vii · Human Anatomy & Physiology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Cranial Nerves & Spinal Nerves
2. Central Nervous System, Input/Output Nerves & Reflex Arc
3. Brain — Cerebrum, Cerebellum & Brainstem Functions
4. Brain & Pain Receptors
5. Cerebrum, Hippocampus & Memory
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1. Cranial Nerves & Spinal Nerves

- Cranium se nikalne wali nerves ko cranial nerves kaha jata hai. Source ke according humans mein cranial nerves ke 12 pairs hote hain aur ye brain ki ventral (bottom) surface par milte hain. [Q1]
- Spinal cord se nikalne wali nerves spinal nerves kehlati hain. Human body mein 31 pairs spinal nerves hote hain; source inhe 8 cervical, 12 thoracic, 5 lumbar, 5 sacral aur 1 coccygeal pair mein group karta hai. Spinal nerves peripheral nervous system ka part hain. [Q2]

2. Central Nervous System, Input/Output Nerves & Reflex Arc

- Source explanation ke mutabik input aur output nerves central nervous system mein, khas taur par spinal cord mein, meet karti hain. Body bhar ki nerves brain ki taraf jaate hue spinal cord mein bundle hoti hain aur reflex arcs bhi spinal cord mein form hote hain. [Q3]
- Reflex arc ek neural pathway hai jo reflex action ko control karta hai. Source ke according higher animals mein most sensory neurons directly brain tak nahi jaate, balki spinal cord mein synapse karte hain; isi wajah se reflex action relatively fast hota hai. Reflex action ka governing centre spinal cord hai. [Q6]
- Source reflex arcs ko do types mein batata hai: autonomic reflex arc (inner organs ko affect karta hai) aur somatic reflex arc (muscles ko affect karta hai). [Q6]

3. Brain — Cerebrum, Cerebellum & Brainstem Functions

- Brain ke three main parts source mein cerebrum, cerebellum aur brainstem diye gaye hain. Cerebrum touch, vision, hearing, speech, reasoning, learning aur fine control of movement jaise higher functions se linked hai. [Q4]
- Cerebellum muscle movements ko coordinate karta hai aur posture/balance maintain karta hai; brainstem breathing, heart rate, body temperature, wake-sleep cycles aur digestion jaise automatic functions perform karta hai. Isliye thinking, heartbeat regulation aur body balance — teeno brain functions ke under aate hain. [Q4, Q5]

4. Brain & Pain Receptors

- Source ke according brain tissue mein pain ya tactile (touch) receptors nahi hote. Isliye needle prick ya brain surgery brain itself mein pain sensation produce nahi karti; given organs mein brain pain feel nahi karega. [Q7]

5. Cerebrum, Hippocampus & Memory

- Memory power ke question mein source correct location cerebrum deta hai. Explanation ke mutabik cerebrum movement, thinking, feeling aur general problem-solving abilities se linked hai, aur hippocampus — jo cerebrum ka part hai — memory se deal karta hai. [Q8]

6. Hypothalamus — Satiety, Thirst & Body Temperature

- Humans mein satiety (satisfaction) aur thirst centres hypothalamus mein located hote hain. Source hypothalamus ko body temperature, thirst, hunger, sleep, circadian rhythm, moods, sex drive aur various hormone release ke regulation se bhi link karta hai. [Q9]
- Body temperature ka regulating centre bhi hypothalamus hai. Source average normal human body temperature 98.6°F batata hai aur hypothalamus ko hunger, thirst, fatigue, sleep aur circadian rhythms ke control se associate karta hai. [Q10]

EXAM TRAP — Hypothalamus vs Other Brain Parts

- ▶ Satiety + thirst centres → Hypothalamus. [Q9]
- ▶ Body temperature regulation → Hypothalamus. [Q10]
- ▶ Balance/posture → Cerebellum; automatic functions such as heart rate → Brainstem. [Q4, Q5]

7. Normal Human Body Temperature

- Source ke according human body temperature winters mein decrease aur summers mein increase nahi hota; normal temperature body thermostat ki tarah hypothalamus ke control mein stable rakha jata hai. [Q11]
- Average normal human body temperature 98.6°F = 37°C = 310 K diya gaya hai. Q12 ke options mein 98.6°F present nahi hai, isliye source correct answer 'None of the above' select karta hai. [Q12]
- Q13 mein normal human body temperature ka correct option 37°C hai; source explanation previous question ki same temperature value refer karti hai. [Q13]

EXAM TRAP — Q12 Options Pattern

- ▶ Normal value source mein 98.6°F hai, lekin Q12 ke options 98.4°F, 98°F aur 98.8°F hain. Isliye source answer (D) None of the above deta hai. [Q12]

8. Medulla Oblongata — Swallowing & Vomiting Reflexes

- Medulla oblongata higher brain levels ko spinal cord se connect karta hai aur autonomic nervous system ke several functions control karta hai. Source ise vomiting, coughing, sneezing aur swallowing ke reflex centre se associate karta hai. [Q14]
- Source pharyngeal reflex, swallowing/palatal reflex aur masseter reflex ko bulbar reflexes ke roop mein mention karta hai. [Q14]